RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251801040-2025-AIDS

B2

Started on	Thursday, 24 September 2026, 10:30 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:38 AM
Time taken	7 mins 56 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int findDuplicate(const int A[], int n) {
    if (n <= 1) return -1;
    int slow = A[0];
    int fast = A[0];
    do {
        slow = A[slow];
        fast = A[A[fast]];
    } while (slow != fast);
    slow = A[0];
    while (slow != fast) {
        slow = A[slow];
        fast = A[fast];
    }
    return slow;
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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